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Seal cartridge with electrically conductive non-woven grounding

Abstract

A cassette seal arrangement for sealing between a shaft and a housing includes a sleeve configured to be mounted to the shaft and including a cylindrical portion and a radially extending portion extending radially outward from the cylindrical portion. A retainer cartridge includes a main retainer configured to be mounted to the housing and supporting a seal member that includes a main seal lip and an exclusionary retainer connected to the main retainer and supporting an exclusionary seal configured to engage the shaft. An electrically conductive non-woven element is disposed longitudinally between the seal member and the exclusionary seal and engages the retainer cartridge and is configured to engage the shaft.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part of U.S. patent application Ser. No. 17/514,986, filed on Oct. 29, 2021. The entire disclosure of the above application is incorporated herein by reference.

FIELD
(1) The present disclosure relates to a seal cartridge with an electrically conductive non-woven element.
BACKGROUND
(2) This section provides background information related to the present disclosure which is not necessarily prior art.
(3) Conventional cassette seals are designed to allow high contamination exclusion by multiple exclusion features.
(4) Current electrically conducting seal arrangements, such as disclosed in U.S. Pat. No. 10,190,690, make a low impedance connection (grounding) between the shaft and housing in order to eliminate electrical interference.
SUMMARY
(5) This section provides a general summary of the disclosure, and is not a comprehensive disclosure of its full scope or all of its features.
(6) A cassette seal arrangement for sealing between a shaft and a housing includes a sleeve configured to be mounted to the shaft and including a cylindrical portion and a radially extending portion extending radially outward from the cylindrical portion. A retainer cartridge includes a main retainer configured to be mounted to the housing and supporting a seal member that includes a main seal lip and an exclusionary retainer connected to the main retainer and supporting an exclusionary seal configured to engage the shaft. An electrically conductive non-woven element is disposed longitudinally between the seal member and the exclusionary seal and engages the retainer cartridge and is configured to engage the shaft.
(7) Further areas of applicability will become apparent from the description provided herein. The description and specific examples in this summary are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

Description

DRAWINGS
(1) The drawings described herein are for illustrative purposes only of selected embodiments and not all possible implementations, and are not intended to limit the scope of the present disclosure.
(2) FIG. **1** is a cross-sectional view of an exemplary cassette seal arrangement according to a first embodiment;
(3) FIG. **2** is a cross-sectional view of an exemplary cassette seal arrangement according to an alternative second embodiment;
(4) FIG. **3** is a plan view of an exemplary electrically conductive non-woven element; and
(5) FIG. **4** is a cross-sectional view of an exemplary cassette seal arrangement according to a third embodiment.
(6) Corresponding reference numerals indicate corresponding parts throughout the several views of the drawings.
DETAILED DESCRIPTION
(7) Example embodiments will now be described more fully with reference to the accompanying drawings.
(8) With reference to FIG. **1**, an exemplary cassette seal arrangement **10** according to the principles of the present disclosure will now be described. The cassette seal arrangement **10** is configured to be mounted between a housing **12** and a shaft **14** for sealing a gap therebetween.
(9) The cassette seal arrangement **10** includes a sleeve **16** configured to be mounted to the shaft **14** and including a first cylindrical portion **16**_a_ that receives the shaft **14** therein and a first radially extending portion **16**_b_ extending radially outward from the cylindrical portion **16**_a_. A second

cylindrical portion **16***c* extends axially from an outer end of the radially extending portion **16***b* and a second radially extending portion **16***d* extending from an end of the second cylindrical portion **16***c*. The sleeve **16** can further include an elastomeric layer **18** on an inner surface of the cylindrical portion **16***a* and on an outboard side of the first radially extending portion **16***b* and the second cylindrical portion **16***c* of the sleeve **16**.

(10) A retainer cartridge **20** is configured to be mounted to the housing **12** and includes a main retainer **22** and an exclusionary retainer **24** connected to one another. The main retainer **22** supports a seal member **26** that includes a main seal lip **28** and a pair of dust lips **30***a*, **30***b*. The main retainer **22** includes a cylindrical outer portion **22***a* that engages the housing **12** and a radially outwardly bent region **22***b* that extends from a first end of the cylindrical outer portion **22***a* and that defines a shoulder that abuts against the housing. The radially outwardly bent region **22***b* terminates in a U-shaped bend **22***c* that captures an outer end **24***a* of the exclusionary retainer **24**. The main retainer **22** includes a first radially inwardly extending portion **22***d* extending inwardly from a second end of the cylindrical outer portion **22***a*. An inner cylindrical portion **22***e* extends axially from a radial inner end of the first radially inwardly extending portion **22***d*. A second radially inwardly extending portion **22***f* extends radially inward from a second end of the inner cylindrical portion **22***e*. The seal member **26** is molded on an inner end of the second radially inwardly extending portion **22***f*. The main seal lip **28** and a first of the pair of dust lips **30***a* engage the cylindrical portion **16***a* of the sleeve **16** while a second dust lip **30***b* of the pair of dust lips engage the first radially extending portion **16***b* of the sleeve **16**. The main seal lip **28** can include a garter spring **32**. The elastomer molded on the second radially inwardly extending portion **22***f* further includes a bumper region opposing the second radially extending portion **16***d* of the sleeve **16**.

(11) The outer end **24***a* of the exclusionary retainer **24** is connected to the U-shaped bend **22***c* of the main retainer **22**. The exclusionary retainer **24** further includes a cylindrical portion **24***b* extending axially from the outer end **24***a* in a direction opposite to the main retainer **22**. A radially inwardly extending portion **24***c* extending radially inwardly from the cylindrical portion **24***b* and supporting an exclusionary seal **36**. The exclusionary seal **36** is configured to engage the shaft and can include a pair of dust lips **36***a*, **36***b*.

(12) An electrically conductive non-woven element **40** is disposed longitudinally between the seal member **26** and the exclusionary seal **36**. The electrically conductive non-woven element **40** has an outer perimeter **40***a* that engages the retainer cartridge **24** and an inner perimeter **40***b* configured to engage the shaft **14**. The electrically conductive non-woven element **40** can be adhered to an outboard face of the first radially extending portion **16***b* of the sleeve **16**. As shown in FIG. **3**, the electrically conductive non-woven element **40** can further include keyhole slots **42** extending radially inward from the outer perimeter **40***a*. The keyholes **42** reduce the radial load of the nonwoven element **40** and reduce the torque required to rotate the shaft **14** and sleeve **16** when in contact with the exclusion/retainer seal.

(13) With reference to FIG. **2** wherein like reference numerals are used to indicate the same or similar elements, an alternative cassette seal arrangement **110** is shown including a modified exclusionary retainer **124** for supporting the exclusionary seal **136** in engagement with an auxiliary sleeve **114** disposed on the shaft **14**. The modified exclusionary retainer **124** has a shorter radially inwardly extending portion **124***c* than the exclusionary retainer **24**.

(14) With reference to FIG. **4** wherein like reference numerals are used to indicate the same or similar elements, an exemplary cassette seal arrangement **210** according to the principles of the present disclosure will now be described. The cassette seal arrangement **210** is configured to be mounted between a housing **12** and a shaft **14** for sealing a gap therebetween.

(15) The cassette seal arrangement **210** includes a sleeve **16** configured to be mounted to the shaft **14** and including a first cylindrical portion **16***a* that receives the shaft **14** therein and a first radially extending portion **16***b* extending radially outward from the cylindrical portion **16***a*. A second cylindrical portion **16***c* extends axially from an outer end of the radially extending portion **16***b* and

a second radially extending portion **16***d* extending from an end of the second cylindrical portion **16***c*. The sleeve **16** can further include an elastomeric layer **18** on an inner surface of the cylindrical portion **16***a* and on an outboard side of the first radially extending portion **16***b* and the second cylindrical portion **16***c* of the sleeve **16**.

(16) A retainer **220** is configured to be mounted to the housing **12**. The retainer **220** supports a seal member **26** that includes a main seal lip **28** and a pair of dust lips **30***a*, **30***b*. The retainer **220** includes a cylindrical outer portion **220***a* that engages the housing **12** and a radially inwardly bent region **220***b* that extends from a first end of the cylindrical outer portion **220***a*. The retainer **220** includes a first radially inwardly extending portion **220***c* extending inwardly from a second end of the cylindrical outer portion **220***a*. An inner cylindrical portion **220***d* extends axially from a radial inner end of the first radially inwardly extending portion **220***c*. A second radially inwardly extending portion **220***e* extends radially inward from a second end of the inner cylindrical portion **220***d*. The seal member **26** is molded on an inner end of the second radially inwardly extending portion **220***e*. The main seal lip **28** and a first of the pair of dust lips **30***a* engage the cylindrical portion **16***a* of the sleeve **16** while a second dust lip **30***b* of the pair of dust lips engage the first radially extending portion **16***b* of the sleeve **16**. The main seal lip **28** can include a garter spring **32**. The elastomer molded on the second radially inwardly extending portion **220***e* further includes a bumper region **34** opposing the second radially extending portion **16***d* of the sleeve **16**.

(17) An electrically conductive non-woven element **240** has a radially outer end **240***a* engaging the cylindrical outer portion **220***a* of the retainer **220** and a radially inner end **240***b* engaging the shaft **14**. The electrically conductive non-woven element **240** can be adhered to an outboard face of the first radially extending portion **16***b* of the sleeve **16**. As noted with respect to FIG. **3**, the electrically conductive non-woven element **240** can further include keyhole slots **42** extending radially inward from the outer perimeter **40***a*.

(18) Example embodiments are provided so that this disclosure will be thorough, and will fully convey the scope to those who are skilled in the art. Numerous specific details are set forth such as examples of specific components, devices, and methods, to provide a thorough understanding of embodiments of the present disclosure. It will be apparent to those skilled in the art that specific details need not be employed, that example embodiments may be embodied in many different forms and that neither should be construed to limit the scope of the disclosure. In some example embodiments, well-known processes, well-known device structures, and well-known technologies are not described in detail.

(19) The terminology used herein is for the purpose of describing particular example embodiments only and is not intended to be limiting. As used herein, the singular forms "a," "an," and "the" may be intended to include the plural forms as well, unless the context clearly indicates otherwise. The terms "comprises," "comprising," "including," and "having," are inclusive and therefore specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. The method steps, processes, and operations described herein are not to be construed as necessarily requiring their performance in the particular order discussed or illustrated, unless specifically identified as an order of performance. It is also to be understood that additional or alternative steps may be employed.

(20) When an element or layer is referred to as being "on," "engaged to," "connected to," or "coupled to" another element or layer, it may be directly on, engaged, connected or coupled to the other element or layer, or intervening elements or layers may be present. In contrast, when an element is referred to as being "directly on," "directly engaged to," "directly connected to," or "directly coupled to" another element or layer, there may be no intervening elements or layers present. Other words used to describe the relationship between elements should be interpreted in a like fashion (e.g., "between" versus "directly between," "adjacent" versus "directly adjacent," etc.). As used herein, the term "and/or" includes any and all combinations of one or more of the

associated listed items.
(21) Although the terms first, second, third, etc. may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, component, region, layer or section from another region, layer or section. Terms such as “first,” “second,” and other numerical terms when used herein do not imply a sequence or order unless clearly indicated by the context. Thus, a first element, component, region, layer or section discussed below could be termed a second element, component, region, layer or section without departing from the teachings of the example embodiments.
(22) Spatially relative terms, such as “inner,” “outer,” “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. Spatially relative terms may be intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the example term “below” can encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

Claims

1. A cassette seal arrangement sealing between a shaft and a housing, comprising: a sleeve mounted on the shaft and including a cylindrical portion and a radially extending portion extending radially outward from the cylindrical portion; a retainer cartridge including a main retainer mounted directly to and makes electrically conductive contact with the housing and the main retainer directly supporting a seal member that includes a main seal lip that contacts the cylindrical portion of the sleeve and an exclusionary retainer formed separate from the main retainer and connected to the main retainer and supporting an exclusionary seal that engages the shaft, wherein the radially extending portion of the sleeve is disposed longitudinally between the seal member and the exclusionary seal; an electrically conductive non-woven element mounted to the radially extending portion of the sleeve and disposed longitudinally between the radially extending portion of the sleeve and the exclusionary seal and having an outer perimeter engaging a cylindrical portion of the exclusionary retainer of the retainer cartridge and an inner perimeter that directly engages the shaft, wherein the electrically conductive non-woven element includes a plurality of slots extending radially inward from an outer perimeter.